

The Blockchain Trilemma

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Bitcoin And Scalability

- Bitcoin was launched in 2009 as the first decentralized digital currency.
- Its goal: build a financial system without central authorities, relying on cryptography and network consensus.
- Bitcoin succeeded: it's secure, transparent, immutable, and not controlled by any single entity.

The Scalability Challenge

- Bitcoin can process only around 7 transactions per second (TPS).
- For comparison, Visa processes thousands of TPS.
- As network usage grows, this limitation becomes a serious issue.

The Visa logo is displayed in white, italicized, sans-serif capital letters on a dark blue rectangular background.

VISA

Why Is Bitcoin Limited?

- Block size is restricted → only a limited number of transactions fit into each block.
- Block creation time is around 10 minutes → new transactions must wait.
- Transaction finality requires several additional blocks → further delays.



What Happens When the Network Is Busy?

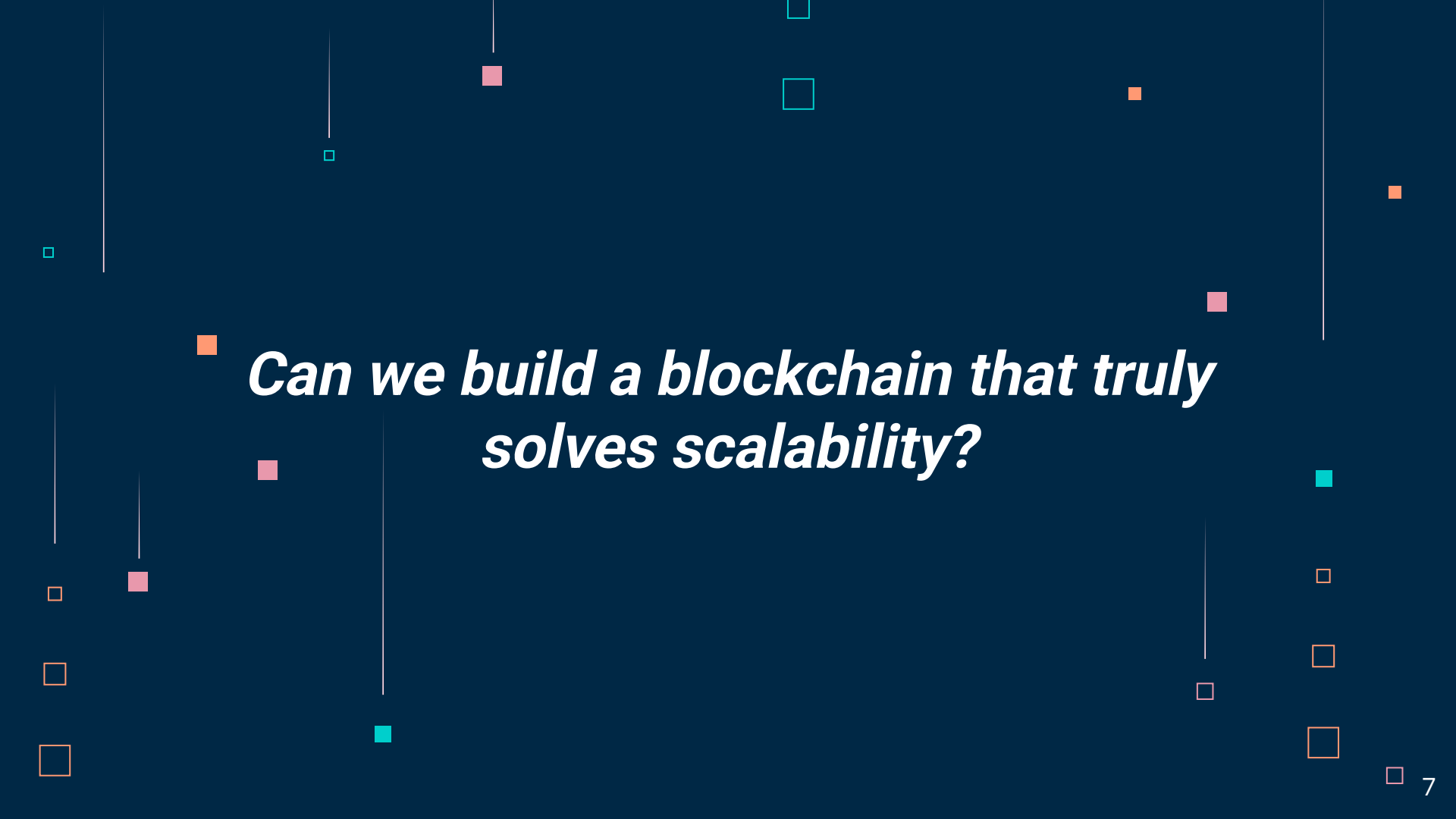
- Transactions queue up.
- Users offer higher fees to get their transactions processed sooner.
- Those who pay lower fees must wait longer.
- Result: higher costs and slower transactions.



The Real-World Impact

- In practical use cases like buying coffee or making instant payments, Bitcoin struggles.
- **Speed and cost are crucial** — and Bitcoin's current design can't always deliver.



The background is a dark blue gradient. It is decorated with a pattern of small, semi-transparent squares in teal, orange, and pink. Some of these squares are connected to thin white vertical lines of varying lengths, creating a sense of depth and movement. The overall aesthetic is modern and minimalist.

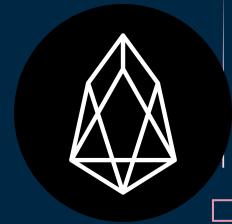
Can we build a blockchain that truly solves scalability?

Fundamental Trade-offs

1. **Ripple**: Centralized consensus algorithm with predefined validators.
Became more like a digital bank model.



2. **EOS**: Block validation assigned to 21 selected nodes.
Created dependency on votes from whales and large companies.



Fundamental Trade-offs

3. **BNB Chain:** Binance's proprietary blockchain with selected validating nodes

Network governance is controlled
by a centralized entity



4. **Solana:** Requires very expensive hardware and resources

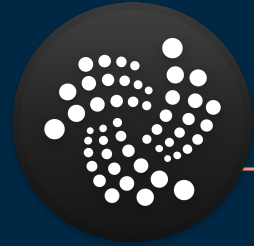
With only a few nodes being able to participate,
reducing public participation



Fundamental Trade-offs

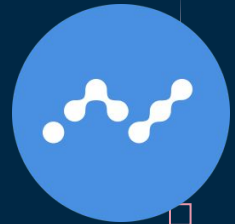
5. **IOTA**: DAG-based architecture aiming for zero fees and high scalability.

Relied on a **central coordinator** for network security



6. **Nano**: Lacks economic incentives for validators.

Making network security dependent on goodwill and **trusted representatives**.

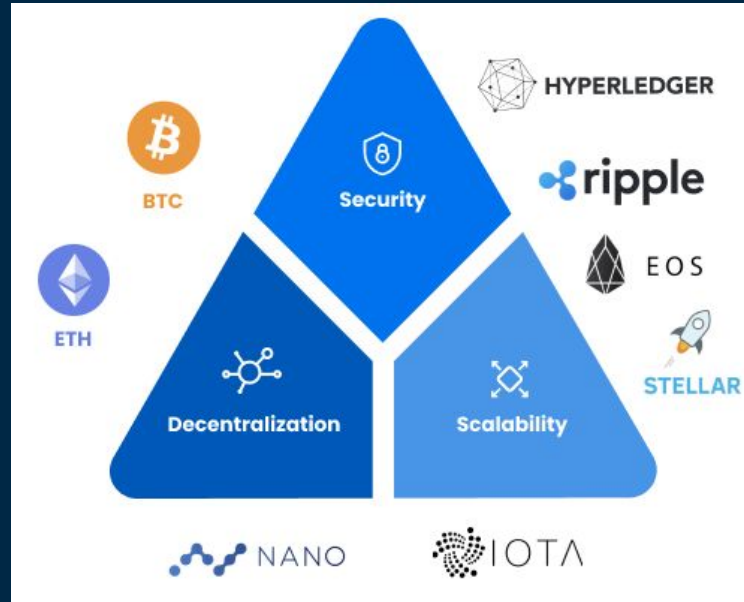


*"It is difficult for blockchains to
■ simultaneously maintain **Security**,
Decentralization, and **Scalability**."*

- Vitalik Buterin -

Blockchain Trilemma

Blockchain projects cannot maintain all three features at the same time.



Security

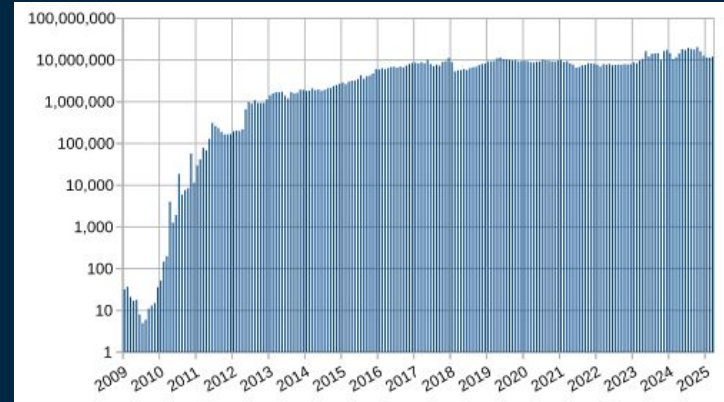
Security is a **top priority** in blockchains. Most networks are **unwilling to compromise security** for **scalability** or **decentralization**, even at the cost of reduced performance or slower speeds.



Scalability

Scalability in blockchain refers to the network's ability to **proportionally increase its transaction processing capacity** as **the number of nodes grows**. Ideally, the number of processed transactions per second would scale as $O(n)$.

In **Bitcoin**, transaction throughput remains around **7 transactions per second** despite a significant **increase** in nodes.



Monthly Number of Bitcoin Transactions

Scalability Solutions

1. **Expensive Hardware:** Utilizing more powerful hardware can **accelerate transaction processing**, but it significantly **increases costs**.
2. **Reducing the Number of Active Nodes:** Limiting the number of active nodes can **improve scalability and speed**, but it compromises **decentralization**.
3. **Changing Blockchain Structure for Better Scalability:** Adopting alternative blockchain structures can **enhance scalability**, but may **weaken security**.

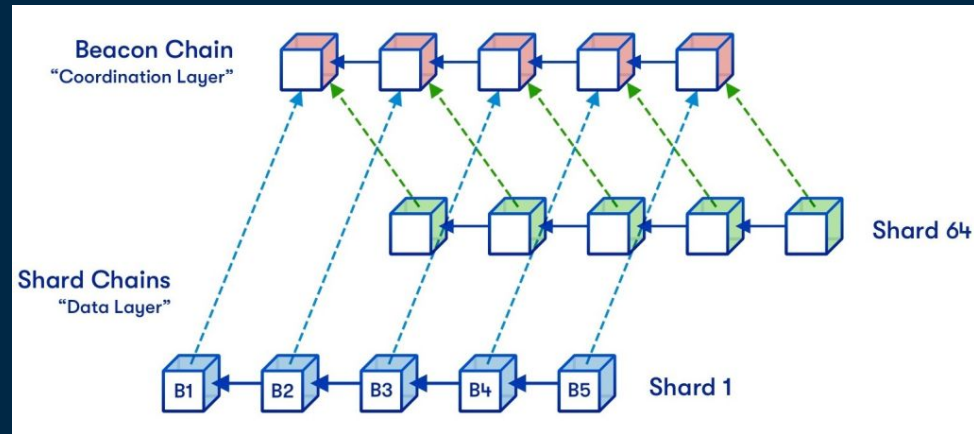
Decentralization

A decentralized network is defined as one in which:

1. No single group controls the network, and no entity fully dominates it.
2. All nodes participate equally, without any node having extra privileges.
3. As the network grows, each node's responsibilities stay the same, with no need for upgrades or added resources.

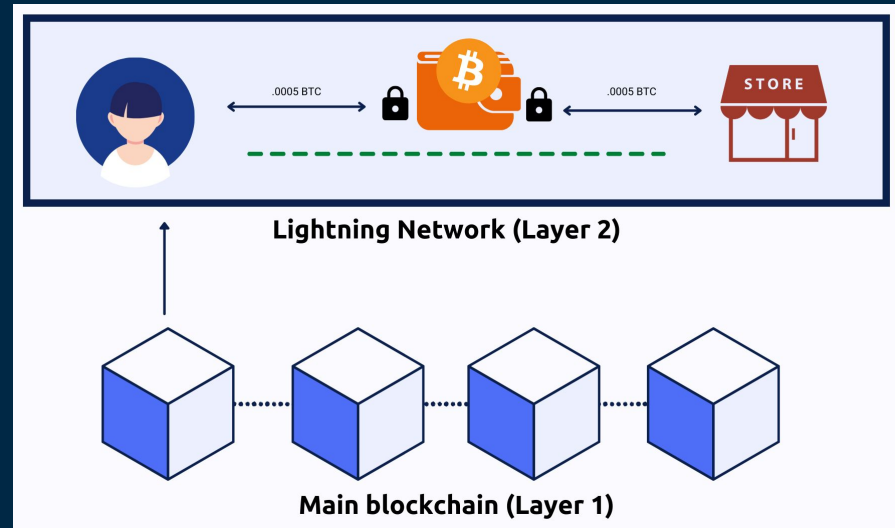
Solutions and Innovations: Sharding

Sharding partitions the blockchain network into smaller sections called shards, enabling **parallel** and **independent transaction processing**. Projects like **Ethereum 2.0** are adopting sharding to significantly **enhance scalability**.



Solutions and Innovations: Layer 2 Solutions

Layer 2 solutions, such as the **Lightning Network (Bitcoin)** and **Polygon (Ethereum)**, operate on top of the main blockchain to reduce **network load**, **lower fees**, and **improve speed** by processing transactions off-chain.



Conclusion

Blockchains cannot simultaneously achieve **security**, **scalability**, and **decentralization**. Solutions like **Sharding** and **Layer 2** improve **scalability**, but often at the cost of compromising **security** and **decentralization**.

The **Blockchain Trilemma** remains **one of the most significant challenges** in the blockchain space. While innovations continue to address this issue, achieving a balance between scalability, security, and decentralization will always be a trade-off, and a definitive solution remains distant.

Resources

- <https://www.coinbase.com/learn/crypto-glossary/what-is-the-blockchain-trilemma>
- <https://www.ledger.com/academy/what-is-the-blockchain-trilemma>
- <https://trustmachines.co/learn/what-is-the-blockchain-trilemma/>
- <https://builtin.com/blockchain/altcoins>
- <https://medium.com/bankless-dao/the-crypto-trilemma-e24d050f6f34>

Any Questions?

Thank you for your attention!